

GUOHAO PANG

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Highlights of Skills

- Expertise in **Internet, wireless networks, mobile networks, AI**;
- **Worked as TA at Portland State University, U.S.A** and familiar with teaching tools such as Canvas;
- Excellent programming skills, including Python, MATLAB, and C++;
- Strong written and verbal communication skills, experience communicating with diverse groups of people;
- Worked for graduate school and familiar with graduate admission systems such as Slate, Banner, and etc.

Education

Texas A&M University College Station, TX PhD Electrical and Computer Engineering	Current GPA:4.0/4.0	(9/2025-present)
Portland State University Portland, OR MS Electrical and Computer Engineering	GPA:3.79	(9/2023-6/2025)
Portland Institute of NJUPT Nanjing, Jiangsu Province, China MS Signals and Communication Systems		(9/2022-6/2025)
Beijing University of Technology Chaoyang District, Beijing, China BS Communications Engineering	GPA:3.55	(9/2018-6/2022)

Publications

- Guohao Pang** and Xi Zhang, "Energy-Efficient Design for Robust UAV Wireless Networks Using Dual-Timescale Statistical Uncertainty Learning," ICC 2026 - IEEE International Conference on Communications, Glasgow, United Kingdom, 2026, pp. 1-6, doi: 10.1109/ICC59461.2026.11586968.
- Guohao Pang** and Xi Zhang, "Energy-Efficient UAV Trajectory and Beamforming Optimizations for mURLLC-Driven AirComp Over ISAC-Enabled Networks," ICC 2026 - IEEE International Conference on Communications, Glasgow, United Kingdom, 2026, pp. 1-6, doi: 10.1109/ICC59461.2026.11588300.
- Guohao Pang** and Xi Zhang, "Resource and Power Optimizations for Multi-Task Air-FL With Digital Twin Assisted ISAC Over OFDM-Based Mobile Networks," ICC 2026 - IEEE International Conference on Communications, Glasgow, United Kingdom, 2026, pp. 1-6, doi: 10.1109/ICC59461.2026.11586876.
- Guohao Pang** and Xi Zhang, "Integrating AirComp With ISAC Over Federated-Learning 6G Mobile Wireless Networks," IEEE International Conference on Communications (ICC) 2025, Montreal, Canada, June 8-12, 2025.
- Guohao Pang** and Xi Zhang, "Federated-Learning Optimized AirComp over UAV-Based CF Massive-MIMO Mobile Networks," IEEE Military Communications Conference (MILCOM) 2025, Los Angeles, USA, October 6-10, 2025.

Guohao Pang, Fengguo Li and Xiaorong Zhu. "Efficiency Optimization Method of Wireless Federated Learning Considering Computational Capability and Channel State," *2023 IEEE 23rd International Conference on Communication Technology (ICCT)*, 2023, pp. 788-793, doi: 10.1109/ICCT59356.2023.10419554.

Xiaorong Zhu & **Guohao Pang** [Accepted]. "A wireless federated learning efficiency improvement method based on joint optimization of collaborative computing and resource allocation". [P]

Guohao Pang & Xiaorong Zhu. "Training Efficiency Optimization Algorithm of Wireless Federated Learning Based on Processor Performance and Network Condition Awareness," *EURASIP Journal on Advances in Signal Processing* 2024, doi: 10.1186/s13634-024-01192-6.

Xiaorong Zhu & **Guohao Pang** [Accepted]. "An Over-the-air Highly Efficient Federated Learning System Integrated Sensing, Communication and Computation and A Federated Learning Method". [P]

Projects

Research on Wireless Mobile Networks and Internet based on ISAC and Cell-free Massive MIMO networks June. 2025-Present

Content: Conducted research on wireless mobile networking architectures integrating Integrated Sensing and Communication (ISAC) with cell-free massive MIMO systems. Focused on joint optimization of communication reliability, sensing accuracy, and network energy efficiency under multi-user and distributed access points deployment scenarios. Developed analytical models and simulation frameworks to evaluate performance trade-offs, including beamforming design, power allocation, and cooperative interference management in large-scale wireless networks.

Influencing Factors of Angle-of-arrival Estimator Based on Deep Learning Jan. 2024-June. 2025

Content: Explores the sensitivity of deep neural networks (DNNs) to different physical conditions and evaluate their adaptability to dynamic channel environments. This thesis also compares the performance variations of traditional estimation methods under the same conditions. These results provide insights into neural network architectures for optimizing AoA estimation under different physical conditions and contribute to developing more robust and versatile deep learning models for wireless positioning and communication systems.

Automatic parking lot car finding system based on UAV Feb. 2020-Oct. 2021

Content: Realize the autonomous patrol flight of UAV indoor. Use of UAV's autonomous cruise technology, three-dimensional reconstruction technology and image recognition technology to realize the detection and recognition of license plate, real-time transmission of image information. Simulate the underground parking lot environment UAV autonomous patrol, image recognition and transmission, as well as three-dimensional reconstruction, the parking space information feedback to the user, and then realize the parking lot automated car-seeking function.

Responsible for UAV flight control module.

Face recognition makeup recommendation system based on deep learning Sept. 2019-Jan. 2020

Content: Use facial recognition to take the color of the face fixed point, the total parameters of the face skin color classification knots a certain law to simulate the effect of the makeup.

Responsible for: Front-end interface programming and function combination.

Experience

Teaching assistant for Microwave Circuit Design I Sept. 2024-Dec. 2025
Electrical and computer engineering department of Portland State University, U.S.A

- Assisted students with problem sets and design assignments on transmission lines, S-parameters, and impedance matching during lab.
- Held office hours and provided guidance on microwave circuit simulations (ADS / MATLAB).
- Graded homework and project reports, and provided constructive technical feedback.

Nucleic acid registrar June. 2022-July. 2022
Neighborhood community

- Maintaining order at the nucleic acid site;
- Assisting physicians in registering nucleic acid records.
- Communicated effectively to diverse stakeholders, such as speaking to patients in more colloquial language and communicating with doctors in a professional tone.

Head of the Sports Department Dec. 2018 –June. 2020
Student Union of Beijing University of Technology

- Organized of sports events and games;
Collaborated with colleagues to develop a schedule for the sports meet and various competitions, including specific time, schedule, sponsorship, rules, and awards. Addressed issues involving conflicts or unexpected events.
- Organized athletes to participate in school level games by issuing notices and contacting capable classmates to participate in the competition.

Volunteer July. 2014–Aug. 2014
International grape conference

- Responsible for guiding visitors to designated locations;
- Responsible for introducing visitors to the attractions in the park.
- Clearly answered questions from Chinese and foreign tourists.
- Quickly located relevant personnel to solve problems.

Awards

“Third prize ” 2020
iCAN International Contest of innovation

“Outstanding Officer” 2020
Student Union, Beijing University of Technology

“Outstanding Class Cadre” 2019
Information Science Department, Beijing University of Technology

“Faculty Scholarships” 2019;
Information Science Department, Beijing University of Technology

“Municipal Outstanding Class Cadre” 2017
Yuying School, Haidian District, Beijing

Other Certifications and Qualifications

- Proficient in office software: Canvas, Zoom, Microsoft Word, Microsoft PowerPoint, Draw.io

- Proficient in video software: Final Cut Pro, Photoshop
- Fluent in English with TOFEL (83)